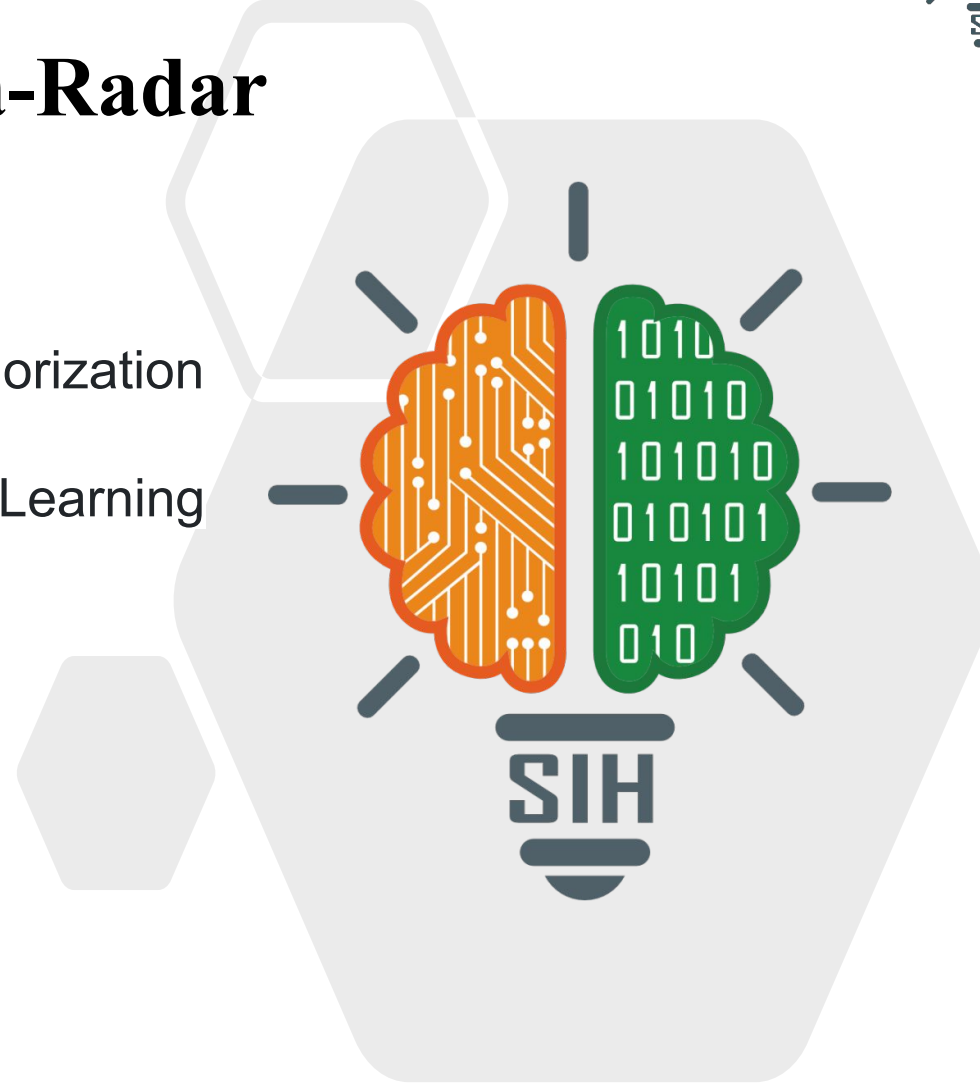


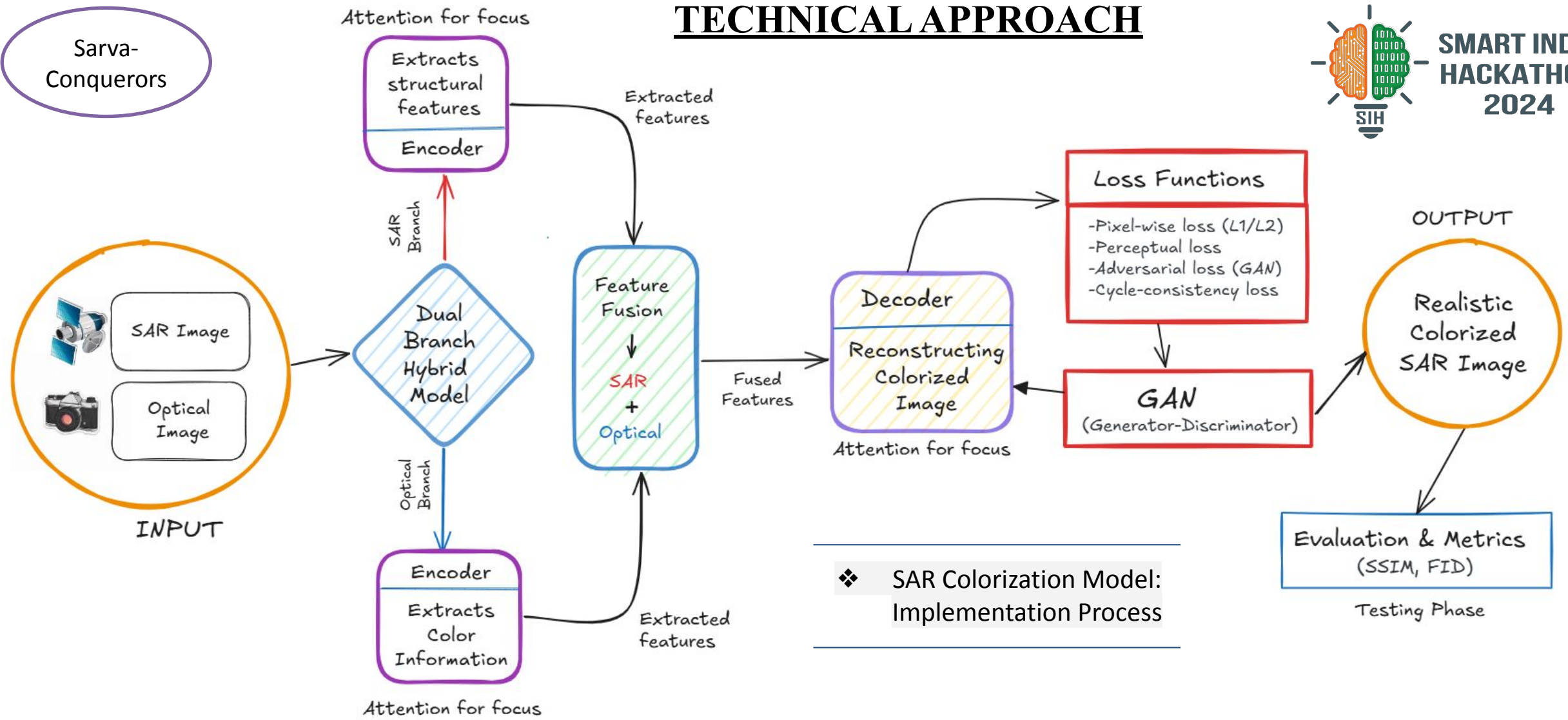
Chroma-Radar

- **Problem Statement ID** – 1733
- **Problem Statement Title**- SAR Image Colorization for Comprehensive Insight using Deep Learning Model (h)
- **Theme**- Space Technology
- **PS Category**- Software
- **Team ID**- 34005
- **Team Name**- Sarva-Conquerors



SAR Image Colorization : Using a Hybrid Model with Attention Mechanisms

- ❖ **Dual-Branch Hybrid Model:** Model uses two branches: one for SAR & one for optical images. Each image passes through its respective **encoder**. The SAR encoder extracts **structural information**, while the optical encoder extracts **color information**. An **attention mechanism** is applied to **focus** on the relevant features of both SAR & optical images.
- ❖ **Feature Fusion:** The features extracted from both branches are **fused** (e.g., through concatenation), integrating the structural information from SAR and color information from optical images to create a unified set of features.
- ❖ **Decoder with Attention:** A decoder **reconstructs the colorized image** from the **fused features**. The attention mechanism in the decoder ensures it focuses on important structural and color cues, ensuring **accurate colorization** of the SAR image.
- ❖ **Loss Functions:** The model uses multiple loss functions to guide training:
 - **Pixel-wise loss (L1/L2):** Ensures the generated colorized image matches the real optical image.
 - **Perceptual and adversarial losses:** Help the model create **visually realistic images**.
 - **Cycle-consistency loss:** Ensures important structural details from the SAR image are preserved during colorization.
- ❖ **Generative Adversarial Network (GAN):**
 - The **Generator** uses the fused features to **produce** the colorized SAR image.
 - The **Discriminator** evaluates realism of generated image **compared** to real optical images, pushing generator to improve.
- ❖ **Final Output: Realistic Colorized SAR Image.**
- ❖ **Evaluation and Metrics:** The model's **performance is evaluated** using metrics. **SSIM** checks how closely the colorized image resembles the real optical image, while **FID** measures the overall visual realism.
- ❖ Applying above techniques together for **realistic SAR image colorization** represents a **unique and innovative approach**.



Technologies
to be used:



- ❖ Is this Idea feasible ?
 - The proposed model is **feasible** due to the availability of paired SAR and optical datasets and the proven success of **hybrid deep learning models** that integrate features from different modalities. The use of **encoders, attention mechanisms, and GAN-based loss functions aligns with state-of-the-art techniques** for generating realistic outputs.
- ❖ Product Offering: We are offering this model to industries like remote sensing, surveillance, environmental monitoring, etc. because it is very useful for them.

Challenges & Risks:

Data Quality (Misalign Images)



Computationally expensive needs
powerful system



Overfitting while Testing



Generalization (Unseen-Data)



Strategies for Overcoming:

Data Preprocessing

Computing Platforms like AWS,
Google cloud, MS Azure, etc

Regularization Techniques

Data Augmentation

Expected Customers:

- **Government agencies** (for surveillance, disaster management)
- **Private companies** (satellite imagery, agriculture)
- **Research institutions** (environmental monitoring, urban planning)

Target Audience - Remote Sensing Scientists, Environmentalists, Urban Planners & Architects, Emergency Responders, Geologists and Geophysicists, Agricultural Managers, Defense and Security Personnel, **General Public**.
(All above people will get benefits for research in there respective fields.)



- **Social Benefits:**

- Better **Communication.**
- Enhances **Decision-making** in many fields.
- Public **Engagement.**



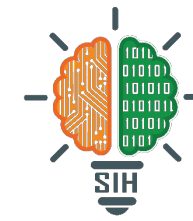
- **Economical Benefits:**

- **Reduces cost** for the need of on-the-ground surveys and inspections.
- Enhances **Monitoring and Management** of resources.



- **Environmental Benefits:**

- **Improves tracking** of environmental changes.
- Sustainable Practices & **Habitat Protection.**
- Effective **Disaster Response.**



- **Feature Extraction in Hybrid Models:**
"A Hybrid Deep Learning Model for Multi-Modal Image Analysis" from [arXiv](#).
- **Dual-Branch Network:** *"Dual-Branch Network for Multi-Modal Image Fusion"*, from [IEEE Xplore](#).
- **IEEE reference:** [SAR Image Colorization Using Multidomain Cycle-Consistency Generative Adversarial Network | IEEE Journals & Magazine | IEEE Xplore](#)
- **KDnuggets:** [Attention mechanism in Deep Learning, Explained - KDnuggets](#)
- **Research Paper For Basic Understanding:**
[A Benchmarking Protocol for SAR Colorization: From Regression to Deep Learning Approaches \(arxiv.org\)](#)
- **Feature Fusion for Multimodal Image Classification**, from [IEEE Xplore](#).
- **Details of Some Loss Functions:** [SAR-to-Optical Image Translation Using SSIM Loss Based Unpaired GAN | IEEE Conference Publication | IEEE Xplore](#)
- **GANs:** [Image Colorization Using Generative Adversarial Networks and Transfer Learning | IEEE Conference Publication | IEEE Xplore](#)

